Tarun Gupta

Data Scientist | Machine Learning Engineer | AI Engineer | Python Developer

Professional Summary

- 10+ years of experience in Data Science, Machine Learning, Deep Learning, Big Data, and MLOps, with extensive Generative AI, LLMs, RAGs, and prompt engineering experience.
- Proficient in deploying AI/ML solutions on AWS, Azure, Snowflake, using CI/CD pipelines (GitHub Actions, Jenkins) and MLOps lifecycle management.
- Strong expertise in Information Retrieval, Data Mining, Computer Vision, CNNs, eXplainable AI, and testing/benchmarking ML/AI systems.
- Skilled in Python, Scala, JavaScript, SQL, HTML/CSS, Java (basics), and frameworks/libraries: Streamlit, Nengo, openCV, Django, Node.js, Spark, Airflow, Tensorflow, Keras, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, Langchain, FastAPI.
- Experienced with cloud platforms, databases, container orchestration, Terraform, Agile methodology, and reporting/visualization (Google Looker, VAPI).

Education

- Bachelor of Technology **||** The LNM Institute of Information Technology, Jaipur, 2016

Certifications

- SnowPro Core Certification
- SnowPro Specialty: Gen AI Certification

Technical Skills

Programming Languages: Python, Scala, JavaScript, SQL, HTML, CSS, Node.js, Java (basics)

Cloud Platforms: AWS (S3, EKS, EMR, Athena, Glue, Lambda, Kinesis, MSK, Sagemaker, SQS, SNS), Azure OpenAI, Snowflake Cortex

Frameworks and Concepts: Streamlit, Nengo, openCV, Django, Node.js, Apache Spark, Apache Airflow, Tensorflow, Keras, PyTorch, Scikit-learn, NumPy, SciPy, Pandas, Matplotlib, Langchain, FastAPI, eXplainable AI

Databases: PostgreSQL, MySQL, MongoDB, Google BigQuery, Snowflake

DevOps / SCM Tools: Git, GitHub, Docker, Jenkins, GitHub Actions, Terraform, Kubernetes

Data Science Skills: Generative AI, LLMs, RAGs, Prompt Engineering, Langchain, CNNs, Machine Learning, Deep Learning, MLOps, Data Mining, Information Retrieval, Computer Vision, eXplainable AI, RNNs, Transformers, NLP

Testing & Benchmarking: Selenium, Mocha, Chai, Confusion Matrix

Reporting / Visualization: Google Looker, VAPI

Other Skills: Web Crawling, Data Analytics, Agile practices, PyCharm

Operating Systems: Unix-based (Ubuntu and macOS)

Spoken Languages: English (Bilingual, C1), German (Beginner, A2), Hindi (Native)

Professional Experience

ATV Tech Solutions | Architect Consultant - Solution Architect - Data Scientist

May 2025 - Present

Project 1: AI Content Machine — Automated Multi-Platform Content Creation System | Industry: Digital Media

Scope: Design and build an end-to-end AI-powered content creation and management system covering ideation, production, repurposing, and multi-platform publishing for a personal brand.

Responsibilities:

- Architected a fully automated content workflow integrating Claude AI and Google APIs for ideation, drafting, and scheduling across YouTube, LinkedIn, Medium, Instagram, Twitter/X, Substack, and podcast platforms.
- Built a multi-agent pipeline using Claude AI to generate scripts, blog posts, and social derivatives from a single content brief.
- Designed Notion-integrated content database to track ideas, production status, and publication history, preventing angle repetition across 90-day windows.
- Implemented analytics ingestion pipeline pulling data from YouTube, Twitter, Instagram, and Medium to feed performance insights back into content strategy.
- Created a reusable prompt library and knowledge base to maintain a consistent creator voice across AI-generated outputs.

Environment: Python, Claude AI (Anthropic), Node.js, Google APIs (YouTube, Search Console), Notion API, Streamlit, n8n, Shell scripting

Project 2: AI Job Hunt Tool — Intelligent Job Search and Application Automation | Industry: HR Tech / Personal Productivity

Scope: Build a comprehensive AI-powered job search automation platform with resume scoring, tailoring, and automated application capabilities across major job portals.

Responsibilities:

- Developed 12 job portal scrapers targeting LinkedIn, Naukri, Greenhouse, Lever, Remotive, WeWorkRemotely, Wellfound, Instahyre, Indeed, Hacker News, Hirist, and ArbeitNow.
- Integrated Gemini Flash for real-time AI-based job scoring against a candidate's resume, ranking opportunities by fit score.
- Built an automated resume tailoring module that dynamically rewrites resume sections to match individual job descriptions.
- Implemented the contact discovery feature to identify recruiter contacts at target companies.
- Designed a Streamlit dashboard for real-time search monitoring, application tracking, and one-click Easy Apply automation via Playwright.
- Built an Excel-based application tracker with full metadata (portal, score, status, contact) for pipeline visibility.

Environment: Python, Streamlit, Playwright, Google Gemini Flash, BeautifulSoup, Selenium, FastAPI, Pandas, Excel automation

Project 3: Voice AI Agents for Customer Service | Industry: Retail/E-commerce

Scope: Implement voice-based AI assistants to handle incoming customer service conversations.

Responsibilities:

- Designed and deployed voice AI agents for autonomous customer interactions using speech-to-text and text-to-speech models.
- Integrated agents with internal systems and external tools for seamless operation.
- Optimized conversation workflows to improve response accuracy and efficiency.

Environment: Python, CrewAI, VAPI, n8n, Streamlit, Twilio, Azure OpenAI, Azure STT, Azure TTS

Project 4: Scalable and Robust ML Pipelines for Multiple Use Cases | Industry: Manufacturing & E-mobility

Scope: Implement end-to-end ML pipelines for multiple use cases, ensuring scalability, robustness, and CI/CD integration.

Responsibilities:

- Designed and implemented dynamic ML pipelines for multiple use cases with dynamic parameterization.
- Built a model registry for versioning and deployment tracking.
- Consolidated datasets for each use case, ensuring clean inputs.
- Integrated pipelines with CI/CD infrastructure for smooth deployment.

Environment: Python, Snowflake, Snowflake Model Registry, GitHub Actions, CI/CD, MLOps

Project 5: Automated KPI Extraction from Documents | Industry: Retail/E-commerce & Manufacturing

Scope: Develop automated pipelines to extract KPIs for various client and prospect documents from diverse file formats efficiently.

Responsibilities:

- Implemented KPI extraction pipelines with dynamic control outside of code with flexible configurations.
- Developed Streamlit dashboards to visualize pipeline outputs and dashboards for monitoring and reporting.
- Mentored team members on pipeline development and dashboard integration.

Environment: Python, Streamlit, Snowflake, Snowflake Cortex, Azure OpenAI, CI/CD, Data Pipelines

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ATV Tech Solutions | Managing Consultant - Lead Data Scientist

April 2023 - April 2025

Project 1: Development of Multiple Custom Data Conversational Chatbots Using GPT-3.5 and Azure OpenAI

Scope: Develop 7 RAG-LLM-based custom data conversational chatbots using GPT-3.5 with Streamlit as the interface using Azure OpenAI.

Responsibilities:

- Integrate GPT-3.5 for sophisticated natural language processing to enhance user interaction.
- Design and build chatbots equipped with advanced feedback mechanisms.
- Integrate suggested follow-up questions to enhance interactive capabilities.
- Implement support for multiple languages to ensure broad accessibility.
- Leveraged Langchain's Conversation Retrieval Agents and Conversation Memories to integrate custom prompts based on client requirements with contextual conversation history.
- Leveraged FAISS from Meta and Cognitive Search from Azure for Vectorstore.
- Integrated Docker and Kubernetes for efficient deployment of machine learning models, automating infrastructure management, and scaling using Terraform.
- Utilized AWS (S3, Lambda) to manage and store chatbot data and CI/CD with GitHub Actions.
- Integrated FastAPI as the API framework to handle chat requests with Azure OpenAI, enabling efficient and scalable API endpoints for real-time conversation data and feedback handling.
- Communicated directly with clients concerning deliverables, requirements, and feedback.
- Supervised and mentored a team of 4, fostering a collaborative and productive work environment.

Environment: Python, Snowflake, Langchain, Azure, OpenAI, Docker, Streamlit, Prompt Engineering, SQL, Generative AI, FastAPI

Project 2: Development of Applications for Invoice Data Extraction and Rule-Based Categorization

Scope: Develop 3 RAG-LLM-based Streamlit applications to fetch information from invoices and categorize rules.

Responsibilities:

- Extract relevant data from various invoices using LLMs by customizing Langchain's Retrieval Agents and their prompts to identify different fields from the documents.
- Store and manage extracted data in the Snowflake database for efficient processing.
- Implement rule-based categorization to classify and organize invoice data.
- Implemented Jenkins for automated CI/CD pipelines, integrating Docker containers for application deployment.
- Used Terraform for infrastructure-as-code, automating AWS resources for efficient application scaling.
- Integrated FastAPI as a lightweight API layer to expose endpoints for fetching, categorizing, and retrieving invoice data.
- Employed FastAPI's async capabilities to handle high concurrency, improving the application's response time for large datasets.
- Ensure data accuracy and consistency through rigorous validation and testing.
- Optimize application performance to handle large volumes of data efficiently.
- Engaged directly with clients to discuss deliverables, gather requirements, and incorporate feedback.
- Managed a team of 3, delegating tasks and responsibilities to optimize team performance and project outcomes.

Environment: Python, Snowflake, Langchain, Azure, OpenAI, Docker, Streamlit, Prompt Engineering, SQL, Generative AI, FastAPI

Project 3: Creation of a Dashboard for Converting Unstructured Text into Structured Output for Emails

Scope: Develop a RAG-LLM-based dashboard to convert unstructured text into structured output for formatted emails.

Responsibilities:

- Integrate the dashboard with GPT-3.5 and Azure OpenAI for advanced text processing.
- Leveraged Langchain's Agents to provide the output email template.
- Design a user-friendly interface for easy input and output of data.
- Automate the formatting of emails based on structured data to improve communication efficiency.
- Utilized FastAPI's support for request validation.
- Leveraged Apache Airflow for task orchestration and scheduling to automate the dashboard's data pipelines.
- Used Apache Spark on AWS EMR for efficient large-scale data processing.
- Provided user support and troubleshoot issues promptly.
- Collaborated directly with clients and business stakeholders to discuss deliverables, gather requirements, and incorporate feedback.

Environment: Python, Snowflake, Langchain, Azure, OpenAI, Docker, Streamlit, Prompt Engineering, SQL, Generative AI, FastAPI

Project 4: Design and Implementation of a Plug-and-Play Environment for Custom Chatbots on Azure OpenAI

Scope: Design a plug-and-play environment for custom chatbots on Azure OpenAI integrated with Snowflake.

Responsibilities:

- Create scalable and customizable chatbot solutions that can easily adapt to different client needs.
- Utilized Kubernetes to manage and orchestrate chatbot deployment, ensuring scalability and reliability.
- Created RESTful APIs using FastAPI to allow seamless integration of custom chatbot components.
- Automated deployment processes with Jenkins and GitHub Actions, using Terraform for infrastructure provisioning.
- Integrate chatbots with Snowflake for efficient data management and processing.
- Provide comprehensive documentation and training to ensure users can effectively utilize the chatbots.

Environment: Python, Snowflake, Langchain, Azure, OpenAI, Docker, Streamlit, Generative AI, FastAPI

Project 5: Development of a Real-Time Human Shape and Gesture Tracking Application Using Computer Vision

Scope: Develop a computer vision application for real-time human shape and gesture tracking.

Responsibilities:

- Design and implement advanced gesture recognition algorithms to track human movements accurately.
- Develop a real-time tracking interface to display and analyze gestures.
- Present the application at WETEX Dubai, showcasing its capabilities and potential applications.

Environment: Python, Streamlit, openCV, SQL

Project 6: Implementation of an Outlier and Anomaly Detection Pipeline with Multiple Machine Learning Algorithms

Scope: Implement an outlier and anomaly detection pipeline with multiple ML algorithms.

Responsibilities:

- Develop and test over 10+ machine learning algorithms to identify outliers and anomalies.
- Handle class imbalance through 15+ different approaches to ensure accurate detection.

- Integrated AWS services (S3, Lambda, SageMaker) to automate the machine learning pipeline.
- Used Docker for containerization and Jenkins for automating the deployment process.
- Integrate the detection pipeline with existing systems for seamless operation.
- Document the pipeline and provide training to users for effective utilization.

Environment: Python, Jupyter notebooks, Streamlit, scikit-learn, SQL

Vivere GmbH | Applied Data Scientist

Hamburg, Germany | May 2022 - February 2023

Project 1: Creation of an LLM-Based Intelligent Dashboard for Identifying New Product Ideas

Scope: Create an LLM-based intelligent dashboard to identify new product ideas.

Responsibilities:

- Utilize Azure OpenAI to enhance product strategy and identify market opportunities.
- Develop the dashboard interface using Streamlit for intuitive user interaction.
- Integrate multiple data sources and analytical tools to provide comprehensive insights.
- Provide user training and support to help stakeholders leverage the dashboard effectively.

Environment: Python, Streamlit, Scikit-learn, PostgreSQL, Google BigQuery, Azure OpenAI, Prompt Engineering, Generative AI

Project 2: Development of an Analytics Dashboard for Marketing Spending Trends and Sales Analysis

Scope: Develop an analytics dashboard for marketing spending trends and sales analysis.

Responsibilities:

- Analyze marketing spending and sales data to identify trends and insights.
- Develop visualizations to show spending trends and sales performance clearly.
- Contrast organic vs. inorganic sales to provide a detailed analysis.
- Optimize dashboard performance to handle large datasets efficiently.
- Provide actionable insights and recommendations to stakeholders based on the analysis.
- Leveraged FastAPI's Pydantic models for data validation, ensuring the accuracy and integrity of the incoming analytics data.

Environment: Python, Streamlit, Pandas, scikit-learn, SQL, FastAPI

Project 3: Implementation of a Competitor Analysis Dashboard Using Statistical Analysis and Clustering

Scope: Implement a dashboard to identify competitors using statistical analysis and clustering.

Responsibilities:

- Analyze competitor data across various marketplaces to identify key competitors.
- Develop clustering algorithms to categorize and group competitors.

- Create visualizations to display the competitor landscape and insights.

Environment: Python, Streamlit, Pandas, scikit-learn, SQL

Project 4: Engineering an Event-Based Notification Service for Email and Asana

Scope: Engineer an event-based email/Asana notification service.

Responsibilities:

- Develop notification triggers to alert users of significant product changes and updates.
- Integrate the notification system with email and Asana for seamless communication.
- Utilized FastAPI to handle API calls for triggering notifications, integrating with external services like email servers and Asana APIs for real-time alerts.

Environment: Python, Email/Asana Integration, FastAPI

MindMorph IT Ventures | Machine Learning Engineer

Remote | January 2019 - March 2022

Project 1: Development of a Human Brain Model (VisMem) to Perform Serial Recall Tasks

Scope: Develop a human brain model (VisMem) to perform serial recall tasks that identify visual inputs from digits similar to human eyes as input sources.

Responsibilities:

- Conduct extensive research on cognitive brain models and their applications.
- Implement and test the functionalities of the brain model to ensure accuracy.
- Analyze the performance of the brain model in various cognitive tasks.
- Document research findings and methodologies for future reference.
- Collaborate with the research team to improve and refine the brain model based on experimental results.

Environment: Python, Nengo, Tensorflow, OpenCV

Project 2: Research on the Effects of Drugs and Injuries on Brain Performance

Scope: Research the effects of drugs and injuries on brain performance while the visual input sources are not up to 100% efficiency.

Responsibilities:

- Analyze the performance of cognitive tasks under different conditions to understand the impact of drugs and injuries.
- Develop methodologies for testing and evaluating brain performance.

Environment: Python, Nengo, Tensorflow, OpenCV

Project 3: Comparison of Query-By-Committee (QBC) and Single Learner Uncertainty for Medical Image Segmentation on Brats2017 Dataset

Scope: Investigated the performance of uncertainty metrics in medical image segmentation, comparing Query-By-Committee with Single Learner Uncertainty.

Responsibilities:

- Explored uncertainty metrics like KLD, JSD, and Entropy to improve informative sample selection by 7% over random sampling for an active learner.
- Demonstrated that Query-By-Committee provides better informative samples with only a 1% fluctuation in precision compared to a 12% fluctuation with single learners.

Environment: Python3, PyTorch, Nibabel, NumPy, SciPy, Matplotlib, MedicalZooPyTorch, modAL, skorch, UNET, VNET

Project 4: Interpretable Deep Reinforcement Learning (eXplainable AI) on MNIST and Space Invaders

Scope: Developed an interpretable deep reinforcement learning model to improve explainability in complex vision-based tasks, focusing on attention visualization in image-based decision-making processes.

Responsibilities:

- Applied deep reinforcement learning (DQN) to computer vision tasks such as MNIST digit classification and Space Invaders, enabling the model to make decisions based on visual input.
- Enhanced model interpretability by generating saliency maps that highlight regions of interest within images, showing where the model focuses during decision-making.
- Improved decision accuracy using Dropout DQN, which demonstrated superior performance over maxpool and unchanged DQN on visual data, leading to better action predictions.
- Increased the DQN performance by 3-6% on the MNIST dataset by selecting representative data points (2.5-10% of the dataset) through MMD-Critic, improving efficiency in visual learning tasks.

Environment: Python3, PyTorch, OpenAI Gym, NumPy, SciPy, FFmpeg, DQN, CNN

Project 5: IRTEX: Image Retrieval with Textual Explanations (eXplainable AI)

Scope: Created an image retrieval system that provides both visual and textual explanations, offering greater transparency and interpretability for users interacting with visual search engines.

Responsibilities:

- Developed an image search engine capable of retrieving images based on user queries and offering both global (set-level) and local (individual image-level) explanations, enhancing user understanding of the search process.
- Used computer vision techniques to extract high-level semantic features, and employed segmentation models to provide detailed visual justifications for retrieved images.
- Provided visual explanations using heatmaps and saliency maps, allowing users to visually comprehend the most relevant parts of the images corresponding to their search queries.
- Achieved high levels of user acceptance (85% for global explanations, 65% for local explanations), demonstrating the effectiveness of combining textual and visual transparency in computer vision tasks.
- Consolidated findings that explainability improves transparency in image retrieval, as users recognized the value of semantic and segmented features with 80% majority approval.

Environment: Python3, Tensorflow, OpenCV, Pandas, NumPy, SciPy, Matplotlib, Django, Angular, MongoDB

Project 6: Development of an Automated Database Migration Framework for MySQL

Scope: Create an automated database migration framework for MySQL.

Responsibilities:

- Develop a comprehensive framework to automate database changes and migrations.
- Test and validate the migration framework to ensure seamless deployment.
- Ensure the framework supports various database schemas and configurations.

Environment: Node.js, Loopback.js, MySQL, RabbitMQ

Project 7: Development of an End-to-End Testing Framework with Extensive Test Coverage

Scope: Develop an end-to-end testing framework with 200+ tests to ensure the reliability of server-side applications.

Responsibilities:

- Implement a comprehensive testing framework to cover all aspects of server-side applications.
- Continuously update and refine the testing framework based on new requirements and feedback.

Environment: Mocha, Chai

Careers360 Pvt. Ltd. | Software Developer

Delhi, India | June 2018 - October 2018

Project 1: Leading a Team for Architecture Design and Development of New Features

Scope: Lead a team for architecture design and development of new features to enhance user experience.

Responsibilities:

- Design and develop new features to improve the user experience on the platform.
- Collaborate with the development team to ensure seamless integration of new features.

Environment: Python, Django, MySQL, HTML, CSS, JavaScript, AWS

Project 2: Development of a Web-Based Chat Application for a 24x7 Doubt Assistance System for Students

Scope: Design and develop a 24x7 doubt assistance system for students.

Responsibilities:

- Develop a web-based chat application to handle student queries and provide assistance.
- Integrate the chat application with existing systems for seamless operation.

Environment: Python, Django, jQuery, AWS

ConveGenius Group | Software Engineer

Noida, India | January 2016 - May 2018

Project 1: Design and Development of a Learning Management System (LMS)

Scope: Design and lead the development of a Learning Management System (LMS).

Responsibilities:

- Manage a team of 3 developers to ensure timely and successful LMS development.
- Ensure the scalability and reliability of the LMS through regular updates and maintenance.
- Collaborate with stakeholders to continuously enhance and improve LMS features based on feedback.

Environment: Django, MySQL, Node.js, Python, AWS

Project 2: Development and Maintenance of Secure and Efficient Node.js APIs for CGSlate Android Applications

Scope: Develop and maintain Node.js APIs for CGSlate Android applications.

Responsibilities:

- Implement secure and efficient backend APIs to support CGSlate Android applications.
- Test and validate API functionalities to ensure reliability and performance.
- Collaborate with the Android development team to ensure seamless integration of APIs.

Environment: Node.js, Express, MySQL, AWS